Relational Algebra Operations

Database as a Set of Relations

- A database is represented as **relations (tables)**.
- Each relation contains multiple **tuples (rows)**.
- Each tuple **must** be **unique**.

Three Core Operators

1. Filtering (Select σ)

- Selects rows based on a condition.
- Operand: One relation.

2. Slicing (Project π)

- Selects specific columns.
- Operand: One relation.

3. Combining (Join ⋈)

- Combines two relations.
- Types of joins:
  - **General Join / Theta Join (⋈θ)** — two operands, condition-based
  - **Equi Join (⋈)** — equality-based
  - **Natural Join (⋈)** — automatically joins on all common attributes

Always ensure the operator and operand are in the correct order.

Example Relation

R (A, B, C)

A	B	C	
1	2	3	
4	5	6	<- all tuples must be DISTINCT
7	8	9	

Selection (σ)

Condition: B = 5 AND A >= 3

A	B	C	→ σ (B=5 AND A>=3)(R)
4	5	6	
7	5	6	

Selection returns only tuples satisfying the condition.

Projection (π)

- Select specific columns (attributes).
- May produce duplicate tuples.

Example: Project C and B from R1

B	C
6	5
6	5

Cartesian Product (×) [CROSS PRODUCT]

T(A, B)

A	B
1	2
3	4
5	6

S (B, C, D)

B	C	D
2	5	6

B	C	D
4	7	8
9	10	11

Result of T × S (9 tuples): (Concatenation)

A	T.B	S.B	C	D
1	2	2	5	6
1	2	4	7	8
1	2	9	10	11
3	4	2	5	6
3	4	4	7	8
3	4	9	10	11
5	6	2	5	6
5	6	4	7	8
5	6	9	10	11

Joins

1. General Join / Theta Join (⋈θ)

Symbol:

```
⋈θ
```

- Joins two relations based on a **condition**.

Example: T ⋈(A >= C) S

GENERAL CONDITION

- T × S
- A >= C

A	T.B	S.B	C	D
5	6	2	5	6

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2. Equi-Cross Product

Symbol:

```
σ (T.B = S.B) (T × S)
```

- Cartesian product filtered by equality.

A	T.B	S.B	C	D		- EQUI CROSS PRODUCT
1	2	2	5	6		- 2 TUPLES
3	4	4	7	8		

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3. Equi Join (⋈)

Symbol:

```
T ⋈(T.B = S.B) S
```

- Combines tuples where specified attributes are equal.

U (A, B, C)

A	B	C
1	2	3
6	7	8
9	7	8

W (B, C, D)

B	C	D
2	3	4
2	3	5
7	8	10

Condition:

- U.B = W.B
- AND

- U.C = W.C

Result:

A	U.B	U.C	W.B	W.C	D
1	2	3	2	3	4
1	2	3	2	3	5
6	7	8	7	8	10
9	7	8	7	8	10

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4. Natural Join (⋈)

Symbol: ⋈

- Automatically joins on all **common attributes**.
- Common columns appear **only once** in the result.

CONDITIONS OF NATURAL JOIN
- EQUI JOIN
- JOINS ON ALL COMMON DOMAINS
- THE COMMON DOMAINS ARE NOT REPEATED IN THE RESULT

Example: U ⋈ W

- U.B = W.B
- AND
- U.C = W.C

Equi-Join:

A	U.B	U.C	W.B	W.C	D
1	2	3	2	3	4
1	2	3	2	3	5
6	7	8	7	8	10
9	7	8	7	8	10

Natural Join:

A	B	C	D
1	2	3	4
1	2	3	5
6	7	8	10
9	7	8	10

- Natural joins remove duplicates automatically.
- Always ensure tuples are distinct.
- Use small, clear examples to visualize joins.
